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Subject: 31261 Internetworking Project

Assessment Task 2: Requirements, Specification and Planning Phase

Project: Encrypted Network Traffic Behavioural Analysis

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1 Introduction

This report provides detailed coverage of the project's objectives, management and methodologies. The aim of this report is to introduce and explain why the project exists, what the project aims to achieve and how it will achieve the set objectives.

1.1 Outline

This document consists of ten sections that provide details of the project. The report includes Introduction, Project Scope and Requirements, Technical Approach and Methodology, Project Plan and Milestones, Roles and Responsibilities, Project Management, Risk Analysis and Mitigation, Ethical and Professional Considerations, Conclusion and References.

1.2 Project Background

With the rapid growth of digital communication and internet-based services, encryption has become a fundamental mechanism for protecting network communications. Encryption protocols such as HTTPS and TLS are widely used to ensure the confidentiality, integrity, and privacy of transmitted data. As a result, a large proportion of modern network traffic is now encrypted.

While encryption protects users and organisations from data interception and eavesdropping, it also introduces significant challenges for network security monitoring. Traditional traffic inspection techniques rely on analysing packet payloads to detect malicious content. However, when traffic is encrypted, the payload data becomes inaccessible to conventional inspection methods, limiting the ability of network administrators and security systems to determine whether traffic is benign or malicious.

One widely used approach for analysing network traffic is Deep Packet Inspection (DPI), which examines packet contents to identify threats. However, DPI is ineffective when traffic is encrypted unless the data is decrypted prior to inspection. Decrypting network traffic introduces several drawbacks, including increased computational overhead, reduced network performance, and potential violations of user privacy. Furthermore, decrypting sensitive communications may raise ethical and legal concerns in many environments.

Due to these limitations, there has been growing interest in alternative approaches that can detect malicious activity without requiring access to packet payloads. One promising direction is the analysis of network traffic behaviour, where statistical characteristics of traffic flows such as packet sizes, timing patterns, and flow duration are analysed instead of packet contents. These behavioural characteristics can reveal patterns associated with malicious activity even when the traffic itself is encrypted.

Recent research has explored the use of machine learning techniques to analyse these behavioural features and automatically classify network traffic. By learning patterns from labelled datasets, machine learning models can identify subtle differences between benign and malicious encrypted traffic flows. This approach enables security monitoring while preserving the confidentiality of encrypted communications.

This project investigates whether behavioural flow features extracted from encrypted network traffic can be used to detect malicious activity through statistical analysis and machine learning techniques, while maintaining a privacy-preserving approach to network security monitoring.

1.3 Research Question

Can behavioural features extracted from encrypted network traffic be used to identify malicious activity without decrypting the packet payload?

1.4 Problem Statement

The increasing adoption of encrypted communication protocols has significantly reduced the effectiveness of traditional network security monitoring techniques that rely on packet payload inspection. As a result, malicious activities such as malware communication and data exfiltration may remain hidden within encrypted channels. Decrypting traffic for inspection introduces performance, privacy, and security concerns, making it an impractical solution in many scenarios. Therefore, there is a need for alternative detection technologies that can detect malicious activity in encrypted network traffic without compromising user privacy or requiring access to the encrypted payload.

This project evaluates whether behavioural flow features such as packet size statistics, timing patterns, and flow duration can provide sufficient information to distinguish malicious encrypted

traffic from benign traffic through statistical analysis and comparative machine learning evaluation.

1.5 Project Objectives

This project aims to investigate the feasibility of detecting malicious encrypted traffic using behavioural analysis and machine learning techniques. The project aims to perform the following:

1. Investigate behavioural characteristics of encrypted network traffic by analysing statistical features such as packet size, flow duration and timing patterns.
2. Perform exploratory data analysis (EDA) to identify patterns and differences between benign and malicious traffic flows.
3. Develop a behavioural traffic analysis framework that processes network flow data and prepares it for machine learning analysis.
4. Train two to three machine learning classification models including Logistic Regressions, Random Forest and/or Support Vector Machine.
5. Evaluate and compare the performance of different models based on accuracy, precision and recall.
6. Identify which behavioural features contribute to the most accurate detection of malicious encrypted traffic.
7. Discuss the practical limitations of using machine learning form encrypted traffic analysis in real-world networks.

2 Project Scope and Requirements

2.1 Project Scope

This section defines the scope of the project, outlining items that are included in the scope and those that are not included.

2.1.1 In Scope Items

The in-scope items for the project are as follows:

- Reviewing existing research papers related to encrypted network traffic analysis and current machine learning approaches used for intrusion detection.

- Selecting a suitable publicly available benchmark dataset, such as the CICIDS2017 for conducting experiments.
- Extracting and analysing behavioural features such as packet size, flow, count etc. from network logs.
- Performing exploratory data analysis to identify patterns between normal and malicious network traffic.
- Implementing machine learning models such as Logistic Regression, SVM and Random Forest for network classification.
- Using appropriate evaluation metrics such as F1-score, precision and recall comparing and evaluate the performance of different models.

2.1.2 Out of Scope Items

The following items are not included in the scope of this project:

- Decrypting encrypted network traffic for training the models.
- Performing deep packet inspection of the network payload.
- Creating a custom dataset by manually simulating different types of attacks.
- Implementing complex deep learning models that require high computational power.
- Developing a real-time intrusion detection system.

2.2 Project Requirements

This section outlines the requirements needed for the successful completion of this project. These requirements include the dataset, tools and resources needed to implement the project.

To conduct the experiments, the project needs a publicly available benchmark dataset that contains accurately labelled normal and malicious network logs. To further enhance the quality of research, the selected dataset should include network traffic that has been exposed to different types of cyber-attacks. Based on these requirements, the CICIDS2017 dataset has been selected for this project because it contains more than 2 million network logs and 14 different types of attacks, making it suitable for our analysis.

Additionally, the project requires a programming environment capable of processing large datasets and implementing various machine learning models. The analysis will be conducted using the

various libraries provided by Python such as pandas, NumPy, Scikit-Learn, Matplotlib etc. that will be used for pre-processing, model implementation and visualisation of results. All the experiments would be carried out in Google Colab and Git will be used for version control and team collaboration.

2.3 Assumptions and Constraints

2.3.1 Assumptions

The following assumptions have been made for this project:

- The selected dataset, CICIDS2017 provides accurately labelled safe and malicious network logs.
- The data for each behavioural feature has sufficient information to perform analysis and compare the results.
- Standard machine learning algorithms are suitable to effectively classify the network logs.

2.3.2 Constraints

The project could be affected by the following constraints:

- **Time:** The project needs to be completed before the final presentation.
- **Dataset:** The project uses publicly available network traffic dataset instead of a real time network traffic dataset.
- **Scope:** The analysis conducted in this project are based on offline and controlled network logs which doesn't fully reflect the unpredictable and complex nature of real-time network traffic.

3 Technical Approach and Methodology

3.1 Literature Review

Almost all the network traffic in today's world is encrypted using protocols such as TLS and HTTPS. While this encryption helps provide privacy and security, it has created a new challenge for the current intrusion detection systems that rely on the packet payloads to detect malicious behaviour. Inspecting encrypted traffic packets isn't easy and requires complex deep packet

inspection techniques which are computational expensive and not ideal for large scale networks. This has led to researchers exploring alternate approaches that focus on detecting malicious behaviour by analysing statistical and behavioural characteristics of encrypted network traffic (Cherukuri, 2024).

Previous research has shown that analysing metadata and flow information from encrypted network traffic can also provide us with useful information without needing to inspect the packet's payload. Behavioural features such as flow duration, packet length and inter-arrival time can be used to identify patterns in the network traffic even when the packet payload is encrypted (Draper-Gil et al., 2016). This proves that behavioural analysis of network traffic is also an effective way to detect malicious behaviour without compromising with the encryption.

One of the most widely used approach to analyse encrypted network traffic is through machine learning models. Research conducted by Pektaş and Acarman (2018) has shown that traditional ML models such as decision trees, random forest and SVMs can successfully differentiate between different types of encrypted logs by using features such as packet sizes and times. Similarly, another study conducted by Anderson and McGrew (2017) has also shown that ML models trained on flow-based features can successfully identify malicious traffic even when the payload remains encrypted.

Along with exploring different detection techniques, the need for a reliable and diverse dataset has been important focus in intrusion detection research. To address the limitations of older datasets, the CICIDS2017 dataset was introduced by Sharafaldin et al. (2018). This dataset addressed the need of traffic diversity, modern attacks and realistic network behaviour in the earlier datasets. Since published, the dataset has been widely used for research and has become a benchmark dataset for intrusion detection.

Overall, existing research has shown that behavioural and statistical features can be used to effectively analyse and classify encrypted network traffic. Many traditional machine learning models can be applied to these features and malicious behaviour can be identified without decrypting the payloads while preserving the privacy and security provided by the encryption protocols.

3.2 Dataset Selection

For this project, the CICIDS2017 dataset has been selected as the primary dataset. This dataset is a benchmark in intrusion detection research and was developed to mimic realistic network traffic, including both benign activity and different type of modern cyberattacks (Sharafaldin et al., 2018).

The CICIDS2017 dataset consists of more than 2 million network logs and includes almost 14 different types of cyberattacks such as Denial of Service, brute-force, botnet activity etc. It also contains a large number of extracted behavioural network features such as packet size, count, inter-arrival times etc. that provide us with the statistics of the network traffic. Along with the features and size, it contains clearly labelled safe and malicious network logs which allows us to effectively train and evaluate different machine learning models.

Overall, the CICIDS2017 dataset provides us with a diverse and well-structured network traffic data that is suitable for training machine learning models and evaluating their performance. It allows us to effectively analyse the behavioural characteristics of encrypted traffic and to compare our results with existing research.

3.3 Methodology

The methodology of this project focuses on analysing behavioural characteristics of encrypted network traffic using machine learning techniques. The objective is to understand if behavioural features can be used to distinguish between normal and malicious activities without decrypting the packet payloads.

The first step involves dataset preparation and pre-processing, For this, the CICIDS2017 dataset will be loaded and analysed to understand its structure and feature distribution, The dataset would be pre-processed by removing missing values, duplicate logs and other unnecessary attributes. To further enhance the performance of the machine learning models, feature scaling and min-max normalisation will also be applied to the dataset.

The next step involves identifying the relevant behavioural features that are needed to classify network traffic. The selected features would mainly be statistical flow characteristics such as flow duration, packet length, count etc. extracted from network logs. These selected features represent the behaviour of the network log and aren't encrypted during communications.

After selecting the appropriate features, an exploratory data analysis would be conducted to better understand the dataset. During the analysis, the statistics of different features will be visually represented using histograms, scatter plots etc. to understand the data distribution, identify patterns, correlations between different features and possibly relationships between different features and the nature of traffic (i.e. normal or malicious network).

After pre-processing and analysing the dataset, machine learning models will be trained to detect malicious traffic. The models that will be used in this project are Logistic Regression, Random Forest and SVM. These models were chosen because of their performance and wide usage in intrusion detection research. The logistic regression model would serve as a baseline for this study because of non-complex nature and easier interpretability. The random forest model will be used to capture more complex non-linear relationships and the SVM model will be used to evaluate margin-based performance classification.

For training and testing purposes, the dataset will be split into 80% training data and 20% testing data. The training dataset will be used to train the models, and the testing dataset will be used to evaluate their performance. Model performance will be evaluated various evaluation metrics would be used, Finally, the results from each model would be analysed and compared to determine which ML model can successfully classify network traffic using only behavioural features.

3.4 Evaluation Metrics

The performance of models will be done using various standard classification metrics. This would help us understand how effectively the models can classify network traffic. The metrics used for evaluation in this project are as follows:

- Accuracy: Accuracy provides the percentage of accurately classified network traffic. It helps in understanding the overall performance of the model. However, it's not reliable with imbalanced datasets which is one of the characteristics of our dataset.
- F1-Score: Addresses the limitation of accuracy and provides us with a balanced measure of the model. It's calculated as the harmonic mean of precision and recall.
- Precision: Shows the proportion of predicted classified malicious traffic.
- Recall: Shows the proportion of malicious traffic that the model identifies.

- Confusion Matrix: Shows the number of TPs, TNs, FPs and FNs by the model. This helps us identify where the model is making errors.
- ROC and AUC Curve: Shows the performance of models across different classification thresholds. This can be used to evaluate the overall performance of the model.

Using different evaluation metrics helps to effectively analyse and understand the model's performance. It also helps us to identify the area of improvements and strengths for each model.

4 Project Management Plan & Milestones

4.1 Communication Plan

The team will carry out written communication on WhatsApp, while online meetings will be carried out on Microsoft Teams. Written communication with the professor will be through emails on Outlook and online meetings via Zoom.

4.2 Meeting Schedule

Meeting with the team will take place one to three times a week through either online and/or in-person meetings. We have agreed with the Supervisor to meet either online or in-person, every second Thursday from 3 to 4 pm to discuss progress and clear off any questions.

4.3 Key Milestones and Deliverables

This project follows an Agile-inspired development methodology with weekly sprints. The work is divided into phases, including Project Planning, Dataset Preparation, Exploratory Data Analysis, Model Development, and Performance Evaluation. Each iteration produces incremental outputs, allowing continuous refinement of the analysis and models throughout the project duration from 5th February to 15th May.

The project is planned within an estimated timeframe where set dates are pessimistic for any sudden changes, incidents or members falling sick. The minimal aim for sprint 10 and 11 is to prepare and work with at least 2 machine learning models, if time allows the project team would prefer to add in the Support Vector Machine to enhance the project accuracy and reliability.

Phases	Sprint	Start Date	End Date	Milestone	Deliverable

Project Planning	**Sprint 1 to 3:** Creating a group & Project idea	05/02	26/02	Research for ideas and find a suitable professor to create a project team.	Project confirmed on ProjectMatch.
	Sprint 4: Research & Solidify project idea	27/02	05/03	Continue research and create a document with details on our project approach.	Project Idea Document.
	Sprint 5: Requirements, Specifications and Planning	06/03	14/03	Project background, scope & requirements, methodology, management plan, risk and mitigation strategies, ethical & professional consideration are detailed in the report.	Requirements, Specifications and Planning Report.
Data Preparation	**Sprint 6:** Data Preparation	15/03	21/03	Identify and download CICIDS2017 dataset, explore dataset structure. Identify behavioural features to prepare dataset for analysis.	Dataset ready for analysis.
Exploratory Data Analysis	**Sprint 7:** Exploratory Data Analysis (EDA) & Comparison	22/03	28/04	Generate statistical summaries of features and compare benign vs malicious traffic.	EDA insights and malicious vs benign identified.
Model Development	**Sprint 8:** Environment Setup	29/03	04/04	Install Python libraries, configure development tools, load dataset into environment.	Software Development Environment.
	Sprint 9: Model Preparation	05/04	11/04	Split the dataset into Testing and Training sets to avoid overfitting.	Dataset ready for models.
	Sprint 10-11: Model Development	12/04	25/04	Implement Logistic Regression & Random Forest Model.	Two trained models.
Performance Evaluation	**Sprint 12:** Model	26/04	02/05	Evaluate models using accuracy,	Models' performance measures.

	Evaluation & Comparison			precision, recall and F1-score.	
	Sprint 13: Final Report	03/05	09/05	Write results, discussion, privacy preserving analysis, finalise report	Project complete.

4.4 Roles & Responsibilities

The project responsibilities are divided across key development phases to ensure systematic progress throughout the project lifecycle. Project responsibilities will be undertaken collaboratively, with all team members actively involved throughout each phase of the project. Although tasks may be distributed among members to ensure efficient progress, the team will maintain a cooperative workflow where members contribute collectively when necessary. Complex or critical tasks may require the participation of the entire team to ensure accuracy and consistency. This flexible structure allows the team to provide mutual support and adjust responsibilities if challenges arise during the project timeline.

Phases	Team Member	Responsibilities
Project Planning	Badia Baghdadi	Define the research problem and project objectives and create project timeline, milestones and sprint plan.
	Prachy Nar	Define the project scope and limitations and develop the project methodology and workflow.
	Saaleha Kadiwala	Create a risk Management and Mitigation Strategy and Identify Ethical and Professional considerations.
Data Preparation	Badia Baghdadi	Perform data cleaning (handle missing values, remove duplicates).
	Prachy Nar	Prepare the dataset for analysis (formatting, labelling).
	Saaleha Kadiwala	Perform feature selection to identify relevant behavioural flow features.
Exploratory Data Analysis	Badia Baghdadi	Generate descriptive statistics (mean, median, variance) and Visuals (histograms, Scatter plot).
	Prachy Nar	Identify behavioural differences between benign and malicious traffic.
	Saaleha Kadiwala	Document insights and observations from the analysis.
	Badia Baghdadi	Prepare the dataset for machine learning modelling.

Model Development	Prachy Nar	Split the dataset into training and testing subsets.
	Saaleha Kadiwala	Train models using the training dataset and store results for evaluation.
Performance Evaluation	Badia Baghdadi	Calculate and compare performance metrics across different machine learning models.
	Prachy Nar	Analyse feature importance and behavioural indicators.
	Saaleha Kadiwala	Interpret results in relation to the research question.

5 Risk Analysis and Mitigation

Priority Rank	Risk Description	Trigger	Probability (L/M/H)	Impact (L/M/H)	Mitigation Response
2	The chosen dataset might not accurately reflect modern encrypted network environments	Realistic encrypted traffic patterns and attack diversity are absent from the dataset.	M	M	Utilise well-known benchmark datasets (CICIDS2017) and make sure the reports explains the dataset's limitations.
1	The size and quantity of features in a dataset might make feature extraction and data preparation difficult.	Inconsistencies of mistakes made when choosing features or cleansing data	M	M	Prior to model training, establish a systemic pretreatment pipeline and verify preprocessing procedures.
3	Machine Learning models might not be able to classify data accurately enough.	Poor performance indicators including recall, accuracy or precision	M	M	Train, evaluate and adjust the parameters of several models (Logistic Regression, Random Forest, SVM).
1	The timeline may be impacted if project tasks take longer than anticipated.	Delays in feature engineering, model training or data analysis	M	H	Create a thorough project timeline with milestones and keep a close eye on developments.

4	Task coordination may be impacted by team members' poor communication.	Missed meetings or ambiguous duties	L	M	Organise frequent project meetings and clearly define roles and duties.
5	An unbalanced dataset could skew predictions and have an impact on model performance.	Significant distinction between malicious and benign traffic samples.	M	M	Use strategies like data balance, sample procedures or assessment measures like F1-score.
6	Computational constraints in Google Colab for handling big datasets	Long processing delays or memory mistakes during model training.	M	M	Utilise features selection, process data in smaller batches and optimise dataset size.
7	Inaccurate interpretation of evaluation measures or model results	Misunderstanding model performance indicators or accuracy	L	M	To validate results, use a variety of evaluation measures, including accuracy, precision, recall and F1- score.
8	Technical problems with tools like version control or python libraries	Git conflicts or library compatibility issues	L	M	Regularly test programs, document dependencies, and uphold version control.
9	Ethical issues with the improper application of cybersecurity research results	The outcomes of the project could be exploited maliciously.	L	M	Make sure research is done solely for academic reasons and that ethical considerations are properly documented.

6 Ethical and Professional Considerations

6.1 Privacy and Data Handling

The goal of this research is to analyse the behavioural traits of encrypted network traffic while protecting user privacy and confidentiality. Deep Packet Inspection (DPI), which needs access to packet payloads in order to identify malicious material, is a common component of traditional network intrusion detection techniques. However, decrypting the data would be necessary to

examine the payload when traffic is encrypted using protocols like HTTPS or TLS. Decryption may violate user privacy and reveal private information.

This research takes a privacy - preserving approach by evaluating the statistical and behavioural characteristics obtained from network traffic flows in order to alleviate these worries. Metadata like packet size distribution, flow duration, packet counts and timing patterns are examples of these properties. The analysis can be carried out without accessing or disclosing sensitive information contained within encrypted packets because these features do not contain the actual contents of the transmission.

Additionally, the CICIDS2017 dataset, which was created for cybersecurity testing, is a publicly accessible research dataset that was utilised in this project. Instead of raw packet contents that can contain personal information, the collection includes network flow records and statistical traffic features. Consequently, private user communications and Personally Identifiable Information (PII) are not included in the dataset.

Every piece of data utilised in this project will be managed carefully and only for scholarly research. Only members of the study team will have access to the dataset and analysis scripts, and all findings will be aggregated and anonymised. These procedures guarantee that the project adheres to ethical research guidelines and protects the confidentiality and integrity of any data used in the analysis.

6.2 Responsible Use of Datasets

The CIDIDS2017 dataset, a publicly accessible benchmark dataset created especially for cybersecurity and intrusion detection research, is used in this project. The dataset contains tagged records of both benign and malicious network activity and was designed to replicate realistic network traffic situations. The dataset offers a solid and well-documented basis for machine learning experiments in network security as it has been extensively used in academic research.

It is crucial to make sure that the data is used appropriately and strictly within the framework of academic study, even though the dataset includes simulated attack scenarios. The dataset's intended used for cybersecurity research will be honoured and it won't be altered in a way that would distort its original structure.

Additionally, when utilising the dataset and any related research articles, appropriate acknowledgement and referencing will be provided. This assures that the research adhered to the standards of academic honestly and transparency. It will also ensure that the contributions of the dataset creators are acknowledged.

Lastly, the initiative will secure accurate and truthful reporting of experimental outcomes. The final report will include a thorough discussion of any dataset limitations, including potential biases and disparities between simulated and real-world traffic conditions. This method helps ensure that the results are understood correctly and encourages ethical research practices.

6.3 Professional Standards

This protects complies with accepted ethical and professional norms in the fields of cybersecurity and information technology. These guidelines place a strong emphasis on user privacy protection, ethical research and reporting, and responsible technology use.

The projects work is in keeping with the guidelines set forth by professional associations like the Association for Computing Machinery (ACM) and the Australian Computer Society (ACS). These organisations support ethical computing principles, such as protecting privacy, guaranteeing equity in technological system, and preventing harm from the improper use of computing technologies.

All research activities will be carried out with integrity, transparency and responsibly throughout the project. To guarantee that the study is repeatable and verifiable, experimental techniques, data processing stages and model evaluation protocols will all the thoroughly recorded. There will also be an upfront acknowledgement of any limitations or uncertainties in the results.

The project team with also uphold professional collaboration techniques, such as open communication, equitable task distribution, and adherence to project deadlines. By adhering to these professional norms, the project will exemplify the ethical behaviour required of information technology professionals and make a responsible contribution to the field of network security.

7 Conclusion

8 Contribution Table

Team Member	Contribution
Badia Baghdadi	• Introduction • Project Plan and Milestones • Project Management • Conclusion
Prachy Nar	• Project Scope and Requirements • Technical Approach and Methodology
Saaleha Kadiwala	• Risk Analysis and Mitigation • Ethical and Professional Considerations • Report Review and Editing

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